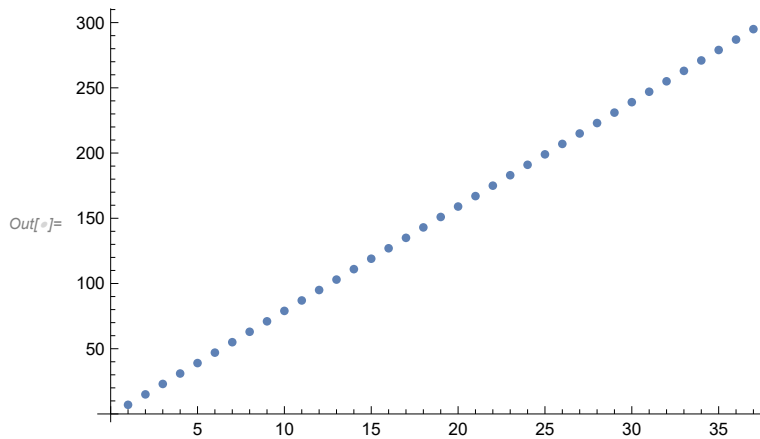

Arithmetic Sequences

$a + b$

```
In[ ]:= Range[7, 300, 8]
```

```
Out[ ]:= {7, 15, 23, 31, 39, 47, 55, 63, 71, 79, 87, 95, 103, 111, 119, 127, 135, 143, 151, 159, 167, 175, 183, 191, 199, 207, 215, 223, 231, 239, 247, 255, 263, 271, 279, 287, 295}
```

```
In[ ]:= ListPlot[Range[7, 300, 8]]
```



These are common, e.g. as a regular way to sample a list uniformly.

Reshaping Arrays

Given a list of size 30,

```
In[ ]:= Range[0, 29]
```

```
Out[ ]:= {0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29}
```

we can reshape the array to a 5x6 grid

```
In[ ]:= Grid[Partition[Range[0, 29], 6]]
```

```
Out[ ]:=
  0  1  2  3  4  5
  6  7  8  9 10 11
12 13 14 15 16 17
18 19 20 21 22 23
24 25 26 27 28 29
```

or to a 6x5 grid

```
In[ ]:= Grid[Partition[Range[0, 29], 5]]
```

```

0  1  2  3  4
5  6  7  8  9
10 11 12 13 14
15 16 17 18 19
20 21 22 23 24
25 26 27 28 29
```

Here the entry in the i -th row and j -th column is $5i+j$

The number n goes in column $j = n \bmod 5$ and in row $i = (n-j)/5 = \lfloor n/5 \rfloor$

We can also make a rank 3 array. Among the choices is 3 by 5 by 2

```
In[ ]:= Grid[Partition[Partition[Range[0, 29], 2], 5]]
```

```

{0, 1} {2, 3} {4, 5} {6, 7} {8, 9}
Out[ ]:= {10, 11} {12, 13} {14, 15} {16, 17} {18, 19}
{20, 21} {22, 23} {24, 25} {26, 27} {28, 29}
```

The possibilities correspond to the factorizations of the number of elements.

Mod

We say x is congruent to y mod m if $x-y$ is divisible by m

$$x = y \bmod m$$

$$x \equiv y \bmod m$$

Reflexive: $x \equiv x$

Symmetric: $x \equiv y$ implies $y \equiv x$

Transitive: $x \equiv y$ and $y \equiv z$ implies $x \equiv z$

Prove that $x_1 \equiv y_1$ and $x_2 \equiv y_2$ implies $x_1 + x_2 \equiv y_1 + y_2$

by showing that the difference is divisible by m .

More complicated is to show that $x_1 x_2 \equiv y_1 y_2$ as long as x, y are integers.

There is an integer k_1 such that $x_1 - y_1 = k_1 m$

There is an integer k_2 such that $x_2 - y_2 = k_2 m$

Then add and subtract:

$$x_1 x_2 - y_1 y_2 = x_1 x_2 - x_1 y_2 + x_1 y_2 - y_1 y_2$$

Now we can factor

$$x_1(x_2 - y_2) + (x_1 - y_1) y_2$$

Plug in the k 's

$$x_1(k_2 m) + (k_1 m) y_2 = m(x_1 k_2 + y_2 k_1)$$

which is a multiple of m because x, y, k are all integers.

This implies for instance $(-1)(-1) = 1$

The main result is that addition, subtraction, multiplication are all well defined mod m .

Every element in the arithmetic sequence $mn + k$ is congruent to k mod m .

These bi-infinite sequences form equivalence classes.

We can think of the addition and multiplication as operations on the equivalence classes.

Power Mod

We want a fast way to calculate $x^y \bmod m$

Use the fact that arithmetic is well-defined mod m , and use mod to keep the numbers small.

The main thing is to reduce the number of operations.

Consider 6^{100} . First consider 6^{64} When the exponent is a power of 2 you can get it by repeated squaring.

$$6^2 \bmod 1000 = 36$$

$$6^4 = 36^2 \bmod 1000 = 1296 = 296$$

$$6^8 = 296^2 \bmod 1000 = 616$$

$$6^{16} = 616^2 \bmod 1000 = 456$$

$$6^{32} = 456^2 \bmod 1000 = 936$$

$$6^{64} = 936^2 \bmod 1000 = 96$$

```
In[ ]:= NestList[Mod[#^2, 1000] &, 6, 6]
```

```
Out[ ]:= {6, 36, 296, 616, 456, 936, 96}
```

Store all the intermediate values because we will use them to compute 6^{100}

Write 100 in base 2, so $100 = 64 + 32 + 4$

$$6^{100} = 6^{64} 6^{32} 6^4 = 96(936)(296) \bmod 1000$$

Repeated squaring gives a fast computation without the mod.

```
In[ ]:= NestList[#^2 &, 6, 6]
```

```
Out[ ]:= {6, 36, 1296, 1679616, 2821109907456, 7958661109946400884391936,
63340286662973277706162286946811886609896461828096}
```

This trick will also work for matrix multiplication. In general whenever an associative operator is iterated.

Divisibility

Assume x and y are not 0. Then x divides y if there is a k such that

$$y = kx$$

Equivalently, x divides y if

$$y \bmod x = 0$$

by the definition of modular equivalence.

The divisors of an integer $x > 0$ are the positive numbers that divide x , including 1 and x

`In[ ]:= Divisors[60]`

`Out[ ]:= {1, 2, 3, 4, 5, 6, 10, 12, 15, 20, 30, 60}`

Primes

A number $n > 1$ is composite if $n = a b$ where $a, b > 1$.

A number $p > 1$ is prime if it is not composite.

That is, there does not exist $1 < x < p$ such that $p = x y$

So a number p is prime if and only if its only divisors are 1 and p

`Out[ ]:=`

n	prime?	divisors
2	prime	{1, 2}
3	prime	{1, 3}
4	composite	{1, 2, 4}
5	prime	{1, 5}
6	composite	{1, 2, 3, 6}
7	prime	{1, 7}
8	composite	{1, 2, 4, 8}
9	composite	{1, 3, 9}
10	composite	{1, 2, 5, 10}
11	prime	{1, 11}
12	composite	{1, 2, 3, 4, 6, 12}
13	prime	{1, 13}
14	composite	{1, 2, 7, 14}
15	composite	{1, 3, 5, 15}
16	composite	{1, 2, 4, 8, 16}
17	prime	{1, 17}
18	composite	{1, 2, 3, 6, 9, 18}
19	prime	{1, 19}
20	composite	{1, 2, 4, 5, 10, 20}

Prime factorization is unique.

One way to find the prime factors is to create a factor tree, e.g.,

$$60 = 20 * 3$$

$$20 = 2 * 10$$

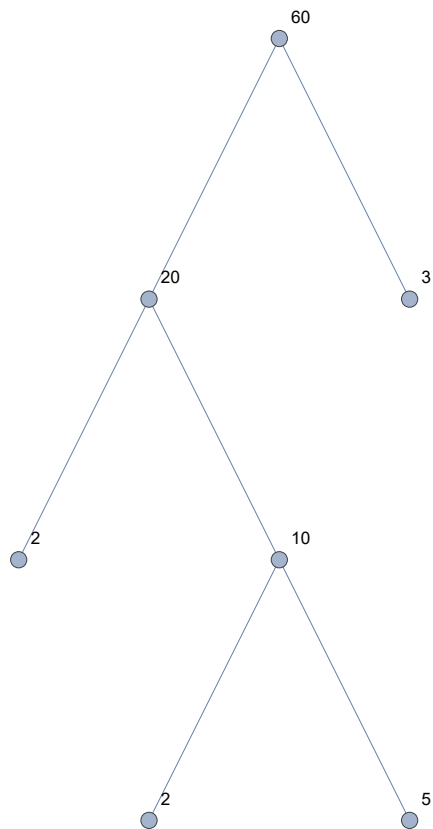
$$10 = 2 * 5$$

```

In[ ]:= TreePlot[{60 → 20, 60 → 3, 20 → 2, 20 → 10, 10 → "2", 10 → 5},
Automatic, 60, VertexLabels → Automatic]

```

Out[]:=

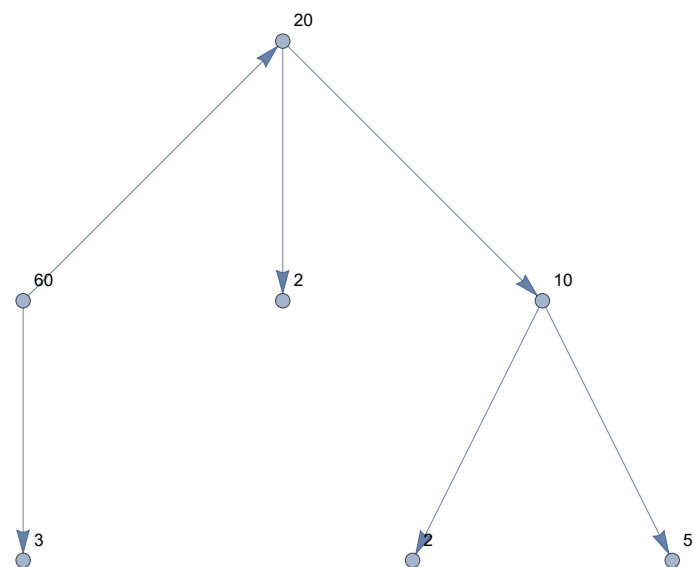


```

In[ ]:= Graph[{60 → 20, 60 → 3, 20 → 2, 20 → 10, 10 → "2", 10 → 5}, VertexLabels → Automatic]

```

Out[]:=



Out[]:=

n	prime?	prime factorization
2	prime	2^1
3	prime	3^1
4	composite	2^2
5	prime	5^1
6	composite	$2^1 \cdot 3^1$
7	prime	7^1
8	composite	2^3
9	composite	3^2
10	composite	$2^1 \cdot 5^1$
11	prime	11^1
12	composite	$2^2 \cdot 3^1$
13	prime	13^1
14	composite	$2^1 \cdot 7^1$
15	composite	$3^1 \cdot 5^1$
16	composite	2^4
17	prime	17^1
18	composite	$2^1 \cdot 3^2$
19	prime	19^1
20	composite	$2^2 \cdot 5^1$

Sometimes we can use prime factorization to solve equations over the integers, and also to show solutions do not exist.

For instance, $x^2 = 2y^2$ has no nonzero solutions.

We can use that to show that $\sqrt{2}$ is irrational.

Relatively Prime

x and y are relatively prime if $\text{GCD}(x,y) = 1$, i.e. they have no common divisors greater than 1.

Facts about GCD

The greatest common divisor can be found by factoring both numbers into primes.

There is a faster way, called the Euclidean Algorithm, which also gives you integers r and s such that

$$\text{GCD}(x, y) = rx + sy$$

Euclidean Algorithm

Find the extended GCD of 123 and 456

Put the big one first

Let $x_0 = 456$ and $x_1 = 123$

Divide and get the remainder.

`In[ ]:= 456 / 123.`

`Out[ ]:= 3.70732`

`In[ ]:= Mod[456, 123]`

`Out[ ]:= 87`

Set $x_2 = 87$

Write the equation: $456 = 3(123) + 87$

as symbols: $x_0 = 3 x_1 + x_2$

Rewrite with x_2 on the left:

$$x_2 = x_0 - 3 x_1$$

Repeat until you get 0.

`In[ ]:= 123 / 87.`

`Out[ ]:= 1.41379`

`In[ ]:= Mod[123, 87]`

`Out[ ]:= 36`

$$x_3 = 36$$

$$x_1 = x_2 + x_3$$

$$x_3 = x_1 - x_2$$

Not zero. Repeat.

`In[ ]:= 87 / 36.`

`Out[ ]:= 2.41667`

`In[ ]:= Mod[87, 36]`

`Out[ ]:= 15`

$$x_4 = 15$$

$$x_2 = 2 x_3 + x_4$$

$$x_4 = x_2 - 2 x_3$$

Not zero. Repeat.

`In[ ]:= 36 / 15.`

`Out[ ]:= 2.4`

`In[ ]:= Mod[36, 15]`

`Out[ ]:= 6`

$$x_5 = 6$$

$$x_3 = 2x_4 + x_5$$

$$x_5 = x_3 - 2x_4$$

Not zero. Repeat.

$$\text{In}[*]:= 15 / 6.$$

$$\text{Out}[*]= 2.5$$

$$\text{In}[*]:= \text{Mod}[15, 6]$$

$$\text{Out}[*]= 3$$

$$x_6 = 3$$

$$x_4 = 2x_5 + x_6$$

$$x_6 = x_4 - 2x_5$$

Not zero. Repeat.

$$\text{In}[*]:= 6 / 3.$$

$$\text{Out}[*]= 2.$$

$$\text{In}[*]:= \text{Mod}[6, 3]$$

$$\text{Out}[*]= 0$$

The remainder is zero so we stop.

The GCD is the last nonzero remainder.

$$\text{GCD} = x_6 = 3$$

Next, we use our equations to find the coefficients of x_0 and x_1 .

$$x_2 = x_0 - 3x_1$$

$$x_3 = x_1 - x_2$$

$$x_4 = x_2 - 2x_3$$

$$x_5 = x_3 - 2x_4$$

$$x_6 = x_4 - 2x_5$$

Substitute.

$$x_2 = x_0 - 3x_1$$

$$x_3 = x_1 - (x_0 - 3x_1) = 4x_1 - x_0$$

$$x_4 = x_2 - 2x_3$$

$$x_5 = x_3 - 2x_4$$

$$x_6 = x_4 - 2x_5$$

$$x_2 = x_0 - 3x_1$$

$$x_3 = 4x_1 - x_0$$

$$x_4 = (x_0 - 3x_1) - 2(4x_1 - x_0) = 3x_0 - 11x_1$$

$$x_5 = x_3 - 2x_4$$

$$x_6 = x_4 - 2 x_5$$

$$x_2 = x_0 - 3 x_1$$

$$x_3 = 4 x_1 - x_0$$

$$x_4 = 3 x_0 - 11 x_1$$

$$x_5 = (4 x_1 - x_0) - 2 (3 x_0 - 11 x_1) = 26 x_1 - 7 x_0$$

$$x_6 = x_4 - 2 x_5$$

$$x_2 = x_0 - 3 x_1$$

$$x_3 = 4 x_1 - x_0$$

$$x_4 = 3 x_0 - 11 x_1$$

$$x_5 = 26 x_1 - 7 x_0$$

$$x_6 = (3 x_0 - 11 x_1) - 2 (26 x_1 - 7 x_0) = 17 x_0 - 63 x_1$$

$$\text{So the GCD} = 17 x_0 - 63 x_1 = 17 (456) - 63 (123)$$

Check

In[6]:= **17 (456) - 63 (123)**

Out[6]= 3

Examples

In[69]:= **Bezout[n_, m_] := NestWhileList[Bezout0, {Sort[{n, m}], "", ""}, FreeQ[0]]**

In[74]:= **Bezout0[{n_, m_}, _, _] :=**
{Mod[m, n], n, Row[{m, " = ", Quotient[m, n], "(", n, ") + ", Mod[m, n]}],
Row[{Mod[m, n], " = ", m, " - ", Quotient[m, n], "(", n, ")"}]}

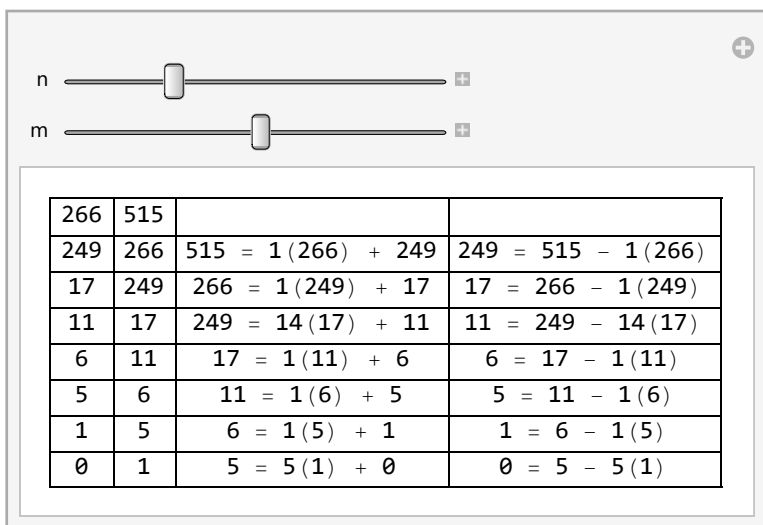
In[75]:= **Grid[Flatten /@ Bezout[45, 69], Frame → All]**

Out[75]=

45	69		
24	45	$69 = 1(45) + 24$	$24 = 69 - 1(45)$
21	24	$45 = 1(24) + 21$	$21 = 45 - 1(24)$
3	21	$24 = 1(21) + 3$	$3 = 24 - 1(21)$
0	3	$21 = 7(3) + 0$	$0 = 21 - 7(3)$

In[73]:= **Manipulate**[Grid[Flatten /@Bezout[n, m], Frame → All], {n, 1, 1000, 1}, {m, 1, 1000, 1}]

Out[73]=



266	515		
249	266	$515 = 1(266) + 249$	$249 = 515 - 1(266)$
17	249	$266 = 1(249) + 17$	$17 = 266 - 1(249)$
11	17	$249 = 14(17) + 11$	$11 = 249 - 14(17)$
6	11	$17 = 1(11) + 6$	$6 = 17 - 1(11)$
5	6	$11 = 1(6) + 5$	$5 = 11 - 1(6)$
1	5	$6 = 1(5) + 1$	$1 = 6 - 1(5)$
0	1	$5 = 5(1) + 0$	$0 = 5 - 5(1)$

In[78]:= **Bezout1a**[n_, m_] := NestWhileList[Bezout1b, {Sort[{n, m}], "", "", 3, {}}, FreeQ[0]]

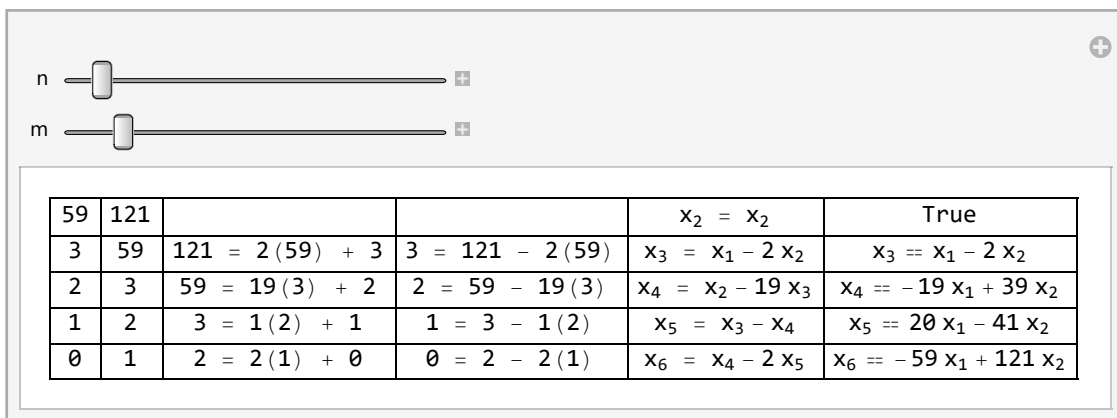
In[94]:= **Bezout1b**[{{n_, m_}, _, _, k_, eqs_] :=

```
{ {Mod[m, n], n}, Row[{m, " = ", Quotient[m, n], "(", n, ") + ", Mod[m, n]}],
  Row[{Mod[m, n], " = ", m, " - ", Quotient[m, n], "(", n, ")"}],
  k + 1, Join[eqs, {x[k - 2] == Quotient[m, n] x[k - 1] + x[k]}]}
```

Out[95]=

266	515			$x_2 = x_2$	True
249	266	$515 = 1(266) + 249$	$249 = 515 - 1(266)$	$x_3 = x_1 - x_2$	$x_3 == x_1 - x_2$
17	249	$266 = 1(249) + 17$	$17 = 266 - 1(249)$	$x_4 = x_2 - x_3$	$x_4 == -x_1 + 2x_2$
11	17	$249 = 14(17) + 11$	$11 = 249 - 14(17)$	$x_5 = x_3 - 14x_4$	$x_5 == 15x_1 - 29x_2$
6	11	$17 = 1(11) + 6$	$6 = 17 - 1(11)$	$x_6 = x_4 - x_5$	$x_6 == -16x_1 + 31x_2$
5	6	$11 = 1(6) + 5$	$5 = 11 - 1(6)$	$x_7 = x_5 - x_6$	$x_7 == 31x_1 - 60x_2$
1	5	$6 = 1(5) + 1$	$1 = 6 - 1(5)$	$x_8 = x_6 - x_7$	$x_8 == -47x_1 + 91x_2$
0	1	$5 = 5(1) + 0$	$0 = 5 - 5(1)$	$x_9 = x_7 - 5x_8$	$x_9 == 266x_1 - 515x_2$

Out[93]=



59	121			$x_2 = x_2$	True
3	59	$121 = 2(59) + 3$	$3 = 121 - 2(59)$	$x_3 = x_1 - 2x_2$	$x_3 == x_1 - 2x_2$
2	3	$59 = 19(3) + 2$	$2 = 59 - 19(3)$	$x_4 = x_2 - 19x_3$	$x_4 == -19x_1 + 39x_2$
1	2	$3 = 1(2) + 1$	$1 = 3 - 1(2)$	$x_5 = x_3 - x_4$	$x_5 == 20x_1 - 41x_2$
0	1	$2 = 2(1) + 0$	$0 = 2 - 2(1)$	$x_6 = x_4 - 2x_5$	$x_6 == -59x_1 + 121x_2$